

Enhancing 5G User-Plane Reliability through Packet-Level Forward Error Correction: A Practical Implementation with Open5GS

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Abstract

Background

The advent of 5G technology has ushered in a new era of ultra-reliable low-latency communications (URLLC), enabling critical applications such as industrial automation, remote surgery, and real-time IoT systems. These applications demand exceptionally high reliability (99.999%) with end-to-end latency under 5ms [1]. Traditional retransmission-based error correction mechanisms like Automatic Repeat Request (ARQ) and Hybrid ARQ (HARQ), while effective in traditional networks, introduce significant latency overhead that becomes problematic for URLLC use cases [2].

State-of-the-Art

Current 5G networks employ HARQ at the MAC layer with sophisticated error correction, but these mechanisms operate exclusively at the radio level. The Open5GS implementation, an open-source 5G core network, provides complete core network functionality but lacks packet-level reliability enhancements at the user plane [3]. Recent research has explored application-layer FEC in various contexts, but there exists a gap in practical implementations that integrate packet-level FEC with real 5G core networks [4].

Issues and Challenges

Several challenges exist in implementing packet-level FEC for 5G user-plane traffic:

- **Radio-Layer Constraints:** Open5GS does not implement PHY/MAC/RLC layers, preventing modifications to native HARQ mechanisms.
- **Latency Overhead:** FEC encoding/decoding must not introduce delays that negate its benefits.
- **Bandwidth Efficiency:** Adding parity overhead must be balanced against network capacity.
- **Integration Complexity:** Seamlessly inserting FEC processing into existing packet flows without modifying core network components.

Objectives

This project aims to design and implement a packet-level Forward Error Correction layer for 5G user-plane traffic flowing through Open5GS. The specific objectives are:

1. Design a non-intrusive FEC proxy architecture that operates on traffic entering and exiting the Open5GS user plane function (UPF).
2. Implement packet-level FEC encoding/decoding using existing libraries (Reed-Solomon and 5G-compliant LDPC codes) integrated into a Python-based processing pipeline.
3. Simulate realistic 5G network impairments including packet loss, delay, and bit errors using Linux tc/netem tools.
4. Evaluate and compare performance metrics including packet loss rate, throughput, latency, and retransmission frequency between baseline (no FEC) and FEC-protected transmission.
5. Analyze the trade-offs between code rate, overhead, and reliability improvement to identify optimal configurations for URLLC scenarios.

The project will provide practical insights into whether packet-level FEC can effectively reduce retransmissions and improve reliability in 5G user-plane traffic, contributing to the broader goal of enabling low-latency critical communications.

Methodology Overview

The proposed system architecture consists of:

- Traffic generator producing controlled UDP flows
- Python-based FEC proxy implementing encoding/decoding
- Open5GS core network as the transmission medium
- Linux tc/netem for network impairment simulation
- Comprehensive measurement framework for performance analysis

Team Details

- **Somyak Priyadarshi Mohanta** (241CS257) - Implementation lead, FEC integration, system architecture
- **Shivam Kumar** (241CS255) - Testing, performance analysis, documentation

References

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